

WAGES IN 1873.

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READ BEFORE THE

SOCIAL SCIENCE ASSOCIATION

AT NORWICH.

BY

THOMAS BRASSEY, M.P.

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LONGMANS, GREEN, AND CO.

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WAGES IN 1873.

Rise in Wages.

IN the following Address I shall devote myself to the task of reviewing, I hope in an impartial spirit, the most recent phases of the labour movement. The great advance of wages is a conspicuous feature of modern English industry, and is obviously due to the rapid growth of the general trade of the country. The long depression following on the panic of 1866 has been succeeded by a period of unprecedented activity in every branch of our export trade. The demands upon the labour-market have far exceeded the supply; and the artisan and labourer have not been slow to take advantage of a situation which afforded to them a brilliant opportunity. Between 1866 and 1869 the value of the exports of the produce and manufactures of the United Kingdom had remained stationary; while it rose from £190,000,000 in 1869, to £256,257,000 in 1872. Such a leap was not possible without imposing a strain upon the powers of our labouring population, which must inevitably have led to a material alteration in the rate of wages.

Not attributable to strikes.

It is too often alleged that the recent advance of wages is attributable to a series of successful strikes. I maintain, on the contrary, that a strike against a falling market is never successful, and that Trades Unions, as an organisation for the purpose of raising wages, can never do more than assist the workman to obtain an advance at a somewhat earlier date than that at which the competition among employers would have brought about the same result. I may quote the unfortunate strike in South Wales, at the commencement of this year, as a signal instance of the inability of Trades Unions to cope with the superior resources of employers when firmly united together.

The recent strike in South Wales.

It may be worth while briefly to recapitulate the most important incidents of the South Wales strike. In June 1872, the miners had proposed to apply for an advance of 20 per cent. on their wages. They were, however, advised by the executive council of their Union to limit their demand to an advance of 10 per cent. The advance was granted, and three months later the men asked for an additional 10 per cent. Their application was refused, and shortly afterwards the masters gave notice of a 10 per cent. reduction. The men thereupon desired that their case should be referred to arbitration. This request was refused by the masters, who were so fully convinced of the strength of their own case, that they offered to submit their books for the inspection of the workmen. The miners were unwilling to avail themselves of this offer; and, encouraged by large promises of support from Mr. Halliday and Mr. Pickard, they went out on strike.

Without venturing to apportion to either of the contending interests their share of responsibility, it is clear that the ironmasters were alone in a position to know whether their business was sufficiently remunerative to make it possible to dispense with a reduction of wages; and it was stated by Mr. Crawshay that he had taken a contract for 2,000 tons of rails at £9. 2s. 5d. per ton nett, and that he lost money by selling rails at that price. Mr. Crawshay expressed an opinion, founded on the statements made by his workmen in daily interviews, that, but for the interference of the Union, they would have been satisfied with the explanations which he had given them, and returned to their work. In short, it became a point of honour with the masters to prove to their workmen that they were able, when acting in concert, to fight a successful campaign against the united forces of the Miners' Union.

The miners, on the other hand, were in the embarrassing position in which workmen are always placed whenever they are engaged in similar disputes. They had to struggle in the dark, and had no means of correctly estimating the profits of their employers. The responsibility of the Executive Council of the Miners' Union, during the labour crisis in South Wales, was immense. Although the miners connected with the Union were only 10,000 in number, by their cessation of labour 50,000 of their fellow-workmen, engaged in various branches of the iron trade, were kept out of work. The

‘strike pay’ distributed by the Colliers’ Union amounted to a total of £40,000, a sum quite insignificant, by comparison with the amount of £800,000, which the men would have earned, had they continued at work; and yet the burden of sustaining a vast population proved eventually insupportable. In point of fact, the men were only enabled to continue the struggle by the assistance of the tradesmen of the district; and when, at length, the latter found themselves unable to continue the supply of the necessities of life on credit, surrender was inevitable.

The reaction against the International Society among the working classes in Belgium originated in a similar cause. In 1871, during the strike in Flanders, the International was unable to fulfil its promises of support, and it has consequently lost credit with the operatives, many of whom, as we are informed by Mr. Kennedy, have withdrawn from the Society. It was the same with the miners at Waldenburg, in Silesia, where 6,000 men went out on strike. After all their savings had been exhausted, they received a grandiloquent despatch from the Central Council at Berlin, urging them to emigrate *en masse*. A few obeyed the advice. The majority who remained were compelled to surrender, being consoled by the assurance that the most valiant armies must sometimes yield to superior numbers, and that they had won for themselves the admiration of Germany.

Almost to the last the originators of the strike in South Wales opposed the generally felt desire to return to work. Never, perhaps, was the magical power of eloquence over an imperfectly educated audience more conspicuously displayed than at the meetings held by the workmen towards the close of the South Wales strike. Men, who had gathered together, for the express purpose of negotiating a peace with their employers, were turned aside against their own judgments by the eloquent exaggerations of orators, who were interested in the continuation of the struggle.

Overwhelming, indeed, is the influence of speech over the uninstructed mind. Well may Carlyle exclaim: ‘He who well considers, will find this same right of speech, as we moderns have it, to be a truly astonishing product of ages; and the longer he considers it, the more astonishing and alarming. I reckon it the saddest of all the curses that now lie heavy on us.’

In the event, as I have said, the workmen returned to their work on the terms, which their masters had originally proposed. Happily they had not long to wait for an improvement of their position; and, in less than a fortnight after the close of the strike, the workmen received an advance of 10 per cent. on the reduced wages, which they had accepted.

Proposal of the International for an universal strike.

The defeat of the miners in South Wales offers, as I have already said, one more illustration of the inability of workmen to force a concession from employers possessed of abundant resources, when the state of trade is such, that a concession cannot be made, without involving the employer in direct pecuniary loss. We have evidence that this fact is becoming generally recognised. The inability of Trades Unions to control the rate of wages was frankly admitted by the members of the International Society in their last congress, when the working men were informed that hereafter, if they wished to secure any substantial advantages for labour, there must be a strike *en masse* of all the working men of every country in the world.

In extreme cases strike may produce results beneficial to workmen. Orderly conduct of men on strike in South Wales.

While I feel bound to assure the working man of the certain frustration of his expectations, if he seeks to obtain from capital impossible concessions, I am at the same time ready to acknowledge that a strike will sometimes make an impression on employers, even in cases, in which the demand for an increase of wages is not immediately conceded. If the trade, in which the workmen on strike are engaged, is prosperous for the employer; cessation of production means loss of profit. The apprehension of a recurrence of such loss may, on a future occasion, induce concessions; and the wage-earning classes may rest assured that, in the long run, and without the assistance of Trades Unions and the disastrous interruptions to their business occasioned by protracted strikes, the competition among employers, to secure the services of workmen, will infallibly lead to a rise of pay, proportionate to the amount of profit, derived from the particular industry, with which they are connected. It was a noteworthy feature in the South Wales strike, that the men never had recourse to physical violence. I

attribute their good conduct in this regard in part to the influence of Mr. Halliday and his colleagues.

Advance in price of coal.

I now pass to the graver subject of the recent rise in the price of coal. It will be remembered that, on the motion of Mr. Mundella, a Committee of the House of Commons was appointed in the last Session to inquire into this subject. After a long investigation the Committee reported, as might have been expected, that, in their judgment, the rapid development of the iron industry was the primary cause of the advance in the price of coal. It appears from statistics, compiled under the direction of the Committee, that the total production of coal in 1869 was 107,000,000 tons, of which 79,000,000 were used in manufactures. The total production in 1871 was 117,000,000 tons, of which 85,000,000 were used in manufactures. It will thus be seen how large a proportion of the total quantity of coal raised is consumed in manufactures, and specially in the manufacture of iron. In 1867, 567,000 tons of pig iron were exported, 4,193,000 tons of pig iron were converted into rolled iron, 1,317,000 tons of rolled iron were exported, and 28,331,000 tons of coal were used in the manufacture of iron. In 1872, 1,333,000 tons of pig iron were exported, 5,390,000 tons of pig iron were converted into rolled iron, 2,055,000 tons of rolled iron were exported, and 38,229,000 tons of coal were consumed in the manufacture of iron.

Mr. Lothian Bell.

In the evidence, which he gave before the Committee, Mr. Lothian Bell stated that the greatly increased demand for the manufacture of iron, although not the sole cause, was one of the causes, of the rapid advance in the price of coal.

In his district the iron trade gave a great stimulus to the coal trade. 'But,' he observed, 'all industry throughout the country has been, and still is, in a very flourishing condition. The manufacture of alkali in the North, the increase of railways, the substitution of steam for sailing vessels, all added to demands on an output not very greatly increasing.' It is to be observed that the rise in the price of iron preceded the rise in the price of coal. Mr.

Lothian Bell quoted figures, from which it appeared that, in September 1871, forge pig iron was selling for 50s., while coke was selling for from 10s. to 12s. a ton. In July 1872, the forge pig iron rose to 120s.—more than double the price of nine months before—and coke, following the advance in iron, rose from 37s. 6d. to 41s. a ton.

The Committee rightly observe, in commenting upon these figures, that, although the disturbance in the proportion between the demand and the supply of coal might not appear sufficient to explain fully the great rise of prices, yet a comparatively small deficiency in the supply of an article of paramount necessity may produce a disproportionate increase of price, through the eagerness of buyers competing with each other, each for his own supply.

Rise of wages in collieries.

Other reasons for the rise in the price of coal have been urged, and among these more especially the reduction in the hours of labour, and the great advance of wages. The advance in the wages paid to miners is in truth extraordinary. In a large colliery, in which I have an interest, I will give the advance in the weekly wages of some of the principal trades. The weekly wages of hewers in 1869 were 24s. 5d.; they have risen in 1873 to 48s. 9d. The wages of timbermen in 1863 were 25s.; in 1873 they are 53s. 4d. Haulers, in 1869, 20s.; in 1873, 31s. 6d. Landers, in 1869, 21s.; in 1873, 36s. 9d. Labourers in 1869, 15s.; in 1873, 24s. a week. The average wages of all the men employed were 20s. 11d. for 1869, as compared with an average of 36s. 8d. per week in 1873.

A similar rise of wages has been established in other parts of the country, of which I have no personal knowledge. Wages have risen, since 1870, 48 per cent. in Northumberland, and 50 per cent. in Durham. The requirements of the Mines Regulation Act have involved an additional expenditure, estimated by some authorities at 12½ per cent. upon the cost of production. It was estimated by Mr. Pease that the total cost of working, in the collieries with which he was connected, had increased 50 per cent. between 1870 and 1872. Mr. A. Macdonald, the president of the Miners' National Association, confirming the opinion of Mr. Pease, estimated that the cost of getting coal

in Northumberland had increased, between 1868 and 1872–73, from 60 to 65 per cent., while the selling price had risen 120 per cent.

Increased profits of colliery proprietors.

It might be easily made to appear that the rise of wages was the principal cause of the advance in coal. But the case would be imperfectly presented for examination, if the profit derived from the working of the pits were not ascertained. The colliery, to which I have already referred, had, for years, been worked at a serious loss—there being no dividend for the proprietors in the years 1870 and 1871. Indeed, the prospects were so gloomy in the latter year, that some of the shareholders in the undertaking made over their interest to their co-proprietors at a considerable discount. At length, however, the tide suddenly turned, and in 1872 an ample dividend was earned; while there is every prospect that the results of the present year may be still more favourable.

Rise of wages has followed advance in price of coal.

My individual experience abundantly confirms the opinion expressed by the Committee of the House of Commons, to the effect that the prices of coal, which prevailed for years before the present rise commenced, were so low that they did not afford a reasonable profit to the owners of collieries in general, or such remuneration as the workmen might, with regard to the hazardous and arduous nature of their labour, reasonably expect. The rise in the rate of wages has not, under the exceptional circumstances, been unreasonable; and it is certain that the real order of events has been, first, the rise in price of iron, then a rise in the price of coal, and lastly a rise in the rate of wages. On the other hand, great as the profits in the coal trade have been, it is a question whether the last two years have compensated the coal-owners for the former protracted era of stagnation, and, in many cases, of serious loss.

In a letter addressed to *The Times*, early in the present year, in which the case of the masters was ably argued, Mr. Laing narrated the history of the Bleanavon Company. Owing to various causes, that concern had been worked for several years without profit. Only within the last three years

had it become a profitable undertaking; and yet all through a long period of adversity an amount of £3,000 to £4,000 a week was paid in wages, at the same rate as by the most prosperous iron works; and the capital sunk by the original proprietors was the means of creating a town, and supporting a population of 9,000, in a secluded mountain valley of South Wales.

Causes which will lead to a reduction in price of coal.

The present unprecedented prosperity may continue for a year or two years at the most, but, at the end of that time, the influx of capital into the coal trade, attracted by the present high profits, will infallibly lead to some reduction of price. New coal pits are being sunk. Old pits are being improved. More workmen are being trained in the business of mining. Hence we may look with confidence to an augmentation of the output, and to a sufficient supply for the ordinary demands of consumers. The insufficient profits of former days cannot be attributed to the unreasonable standard at which wages were maintained. The excessive competition in the supply of coal was the true cause of the unfortunate position of the trade. And as in the former period of depression, so in the sudden and it may be short-lived prosperity of the present day, the rates of wages must be regarded, not as a cause, but as a consequence, of an abnormal position of affairs.

Hours of labour.

Complaints have been urged as to the effects of shortening the hours of labour; and it is certain that if a comparison be made between the amount raised and the total number of individuals employed, a less quantity is raised than in former years. It must not, however, be forgotten that high wages have attracted many untrained hands to the coal pits. It would be presumptuous in me to express an opinion as to the precise number of hours, which would constitute a fair working day in a coal pit. Mr. Macdonald, who has had actual experience as a working miner, declares that the present earnings could not be obtained with less than eight hours of work a day, and that no man, who laboured assiduously for that number of hours could work continuously six days a week at coal mining. It will be the duty of those, to whom the miners are in the habit of looking for

guidance, to watch with care the course of trade. They know that the iron manufactures of this country can only prosper, so long as we are able to sell our iron abroad at cheaper rates than those demanded by foreign producers.

There are some who think that a limitation of the hours of labour is in itself an evil. I cannot share in this view. Because some may make an unwise use of their newly acquired advantages, that is no reason for returning to a former state of things; when, in the general depression of trade, an undue pressure was brought to bear upon the working man. 'No doubt,' says Sir Arthur Helps, 'hard work is a great police agent. If everybody were worked from morning till night, and then carefully locked up, the register of crime might be greatly diminished. But what would become of human nature? Where would be the room for growth in such a system of things?'

The use of leisure requires education, and that education had not been freely given to the mechanics, miners, and puddlers, of former generations.

The double-shift system.

Among the various proposals for maintaining the production of collieries while conceding to the individual workman the advantage of a reduction in the number of hours of daily labour, the double-shift system of working promises a most satisfactory result. When the memorable struggle was commenced in Newcastle for a reduction in the number of hours, I ventured to suggest, in an address delivered at Birkenhead, that the solution of the difficulty, which had arisen in the engineering establishments, might be found in employing relays of mechanics to succeed each other at the same machine. When we have to combine human labour with machine power, we know that it is impossible for the human machine to keep pace with machinery of brass and iron. But why, I venture to ask, should not a machine, which never tires, be tended by two or three artisans relieving each other as one watch relieves another on board ship? In driving the machinery of steamships, it has been found necessary, on long voyages, to have three sets of engineers and firemen. Why should not the day be divided into three periods of eight hours, or the working day be extended to sixteen hours, two sets of men being employed? The change, arising from the increasing use of machinery, seems to render corresponding modifications in the application of labour essential. My friend Mr. Elliott is

pushing the system of a succession of labour in collieries with very advantageous results to all parties concerned. Comparing a Durham colliery, worked on the double-shift system, with a colliery in Glamorganshire, worked by one set of miners, he ascertained that twice the quantity of coal per day was being raised in Durham. The prejudices of the miners in South Wales against the double-shift have presented a serious obstacle to its introduction, but Mr. Elliott hopes that this may eventually be overcome by the influence of Mr. Macdonald and other representatives of the men, whose superior intelligence will enable them to appreciate more readily the advantages of new and improved systems of working.

Coal-cutting machines.

Among various improvements, which may tend to reduce the price of coal, we may look with confidence to the increased use of coal-cutting machinery as a substitute for manual labour; and to the discovery of methods by which the consumption of fuel may be reduced. The experiments, which have been tried with the machines invented by Captain Beaumont, R.E., and others, have been eminently satisfactory; and these machines are now being made in large numbers in Glasgow and Birmingham.

Waste of coal in domestic consumption.

Our domestic consumption is undoubtedly wasteful; and the inventor of an effective improvement in the form of grate in common use will be a real benefactor to his fellow-man. Already we have, in the cooking-stove for yachts, the invention of Mr. Atkey, of Cowes, a highly successful apparatus. A letter from Mr. Vale, Ex-President of the Liverpool Architectural Society, addressed to *The Times* in August last, describes a cooking stove for a party of nine persons and a crew of thirteen men, which measured only one foot four inches by one foot four inches in area, and one foot nine inches in height, the actual fuel-space being less than one cubic foot. The fuel required in his yacht for one day's consumption was forty-seven pounds of coke at twenty shillings a ton, and the cost per head per day amounted to less than one farthing.

Captain Galton's fireplace.

In his lecture, delivered at Bradford during the meeting of the British Association in the present year, Mr. Siemens described Captain Galton's ventilating fireplace as a most valuable invention.

'The chief novelty and merit,' he said, 'of Captain Galton's fireplace consists in providing a chamber at the back of the grate, into which air passes directly from without, becomes moderately heated (to 84° Fah.), and, rising in a separate flue, is injected into the room under the ceiling with a force due to the heated ascending flue. A plenum of pressure is thus established within the room whereby indraughts through doors and windows are avoided, and the air is continually renewed by passing away through the fireplace chimney as usual. Thus the cheerfulness of an open fire, the comfort of a room filled with fresh but moderately warmed air, and great economy of fuel, are happily combined with unquestionable efficiency and simplicity; and yet this grate is little used, although it has been fully described in papers communicated by Captain Galton, and in an elaborate report made by General Morin, le Directeur du Conservatoire des Arts et Métiers of Paris, which has also appeared in the English language.'

Mr. Bessemer's inventions.

But economy in the consumption of coal, in the manufacture of iron, is a far larger question than economy, however desirable in itself, in the consumption for domestic purposes; and, as an illustration of what may be achieved in this direction, I will quote some extracts from a letter from Mr. Bessemer, detailing the results, which have actually been attained through his most valuable discoveries.

The average quantity of coal required to make a ton of pig iron is about two tons of coal to a ton of pig; and, as pig iron forms the raw material for the several processes of manufacturing both malleable iron and steel, we may treat the pig simply as the raw material employed, and consider only how much coal is required to make a ton of finished rails. About two tons of coal are required in order to convert pig iron into iron railway bars.

To produce one ton of steel rails by the old process of making steel in Sheffield, a total consumption of ten tons eight cwt. of coal is required; and the conversion of iron bars into blistered bars occupies from 18 to 20 days.

To make Bessemer's steel from pig iron into steel rails requires about five cwt. of small coal, in the form of coke, to melt the pig iron in the cupola; two cwt. to heat the converting vessel and ladle; two cwt. for the blast engine, which converts five tons of pig iron into fluid cast steel in twenty minutes; and, lastly, for rolling the ingots into rails, sixteen cwt. of coal, making a total consumption of twenty-five cwt. of coal, in producing one ton of Bessemer's steel rails from pig iron. Thus, common iron rails take two tons of coal; Sheffield cast steel rails, ten tons eight cwt.; Bessemer's steel rails, one ton five cwt.

But we must also consider other points in connection with these figures, in order to arrive at a correct estimate of the saving of coals, effected by the introduction of steel, as a substitute for iron.

Although the cost of Sheffield steel entirely shut it out of the market for rails, it must be borne in mind that it was extensively used for wheel tires, slide bars, piston rods, and other parts of locomotive engines; and here a saving of over nine tons of coal per ton of steel has been effected. Further, it must be borne in mind that at stations where rails are rapidly worn, the saving by the use of steel, as a substitute for iron, must not be simply estimated as a saving made on one ton of each material. For instance, at the London and North-Western station, at Crewe, the iron rails are so rapidly worn, that they require to be reversed every four months, each rail being completely worn out in eight months. Bessemer's steel rails were first used at this station, and after being in constant use for seven years, they were removed in consequence of rebuilding the station; one side only of the rail having been used, and this was not quite worn out. During the seven years, therefore, that those rails were down, one ton five cwt. only of coal had been employed in the production of each ton of rails used at this station; whereas ten sets of iron rails would have been entirely worn out in that period, each set consuming two tons of coals in its manufacture, or equal to twenty tons of coals for iron rails, as against one ton five cwt. of coals for steel rails; and these, when turned, would be equal to another seven years' wear on the side not used.

The above is, no doubt, an extreme case, but the same sort of thing goes on everywhere where steel is used, though in a lesser degree. It has indeed been admitted by competent persons, that the rapid destruction of iron rails

would have caused a complete collapse of the Metropolitan railways by continued interference with the traffic, while removing the worn-out rails, had not steel been employed.

It should further be borne in mind that the extra strength of steel over iron admits of a reduction of one-third of its weight in all structures, previously made in iron. Thus, a further saving is effected in the fuel consumed for a given work.

The rapidity, with which Bessemer's steel is coming into use, will be appreciated, when it is stated that the report of the jury at the London International Exhibition showed that the entire production of steel in Great Britain, prior to Bessemer's invention, amounted to 51,000 tons per annum; while the quantity of Bessemer's steel, made in Great Britain during the twelve months ending June 1873, amounted to 481,000 tons, or nearly ten times the amount of production prior to the invention. Had this quantity of steel been made by the old Sheffield process, it would have consumed, according to the foregoing figures, 4,401,000 tons more coal than was actually employed in its production. Should this enormous increase in the manufacture continue, as it at present promises to do, in another five years, we may have treble the quantity of steel made in this country with a corresponding saving of fuel.

Reduced consumption of coal in steamers.

In steam vessels a remarkable economy of fuel has of late been attained. In his lecture at Bradford, Mr. Siemens said, 'A striking illustration of what can be accomplished in a short space of time was brought to light by the Institute of Mechanical Engineers, over which I have at present the honour to preside. In holding their annual general meeting in Liverpool in 1863, they instituted a careful inquiry into the consumption of coal by the best engines in the Atlantic steam service, and the result showed that it fell in no case below 4½ lbs. per indicated horse-power per hour. Last year they again assembled with the same object in view in Liverpool, and Mr. Bramwell produced a table showing that the average consumption by 17 good examples of compound expansive engines did not exceed 2¼ lbs. per indicated horse-power per hour. Mr. E. A. Cowper has proved a consumption as low as 1½ lbs. per indicated horse-power per hour in a

compound marine engine, constructed by him with an intermediate superheating vessel. Nor are we likely to stop long at this point of comparative perfection, for in the early portion of my address I have endeavoured to prove that theoretical perfection would only be attained if an indicated horse-power were produced with 1/5.5 lb. of pure carbon, or say ¼ lb. of ordinary steam coal per hour.'

Inventions of Mr. Siemens.

The furnace invented by the Messrs. Siemens is another highly successful contrivance. In melting one ton of steel in pots, 2½ tons of coke are ordinarily consumed. In Messrs. Siemens' furnace, a ton of steel is melted with 12 cwt. of small coal.

When such results as this have been secured by a few inventors, what may we not venture to expect from the concentration of many ingenious minds on the important problem of economising coal?

Substitution of peat for coal.

As it is, I fear, certain that coal will never fall below fifteen or sixteen shillings a ton at the pit's mouth, we ought not to neglect any possible source of supply. It has been suggested that the time has come when the peat bogs of these islands should be turned into coal and charcoal; and I am informed that the cost of the process does not exceed five shillings a ton. If these anticipations are verified, the drain on our coal-pits will be materially lessened.

General prospects of the iron trade.

It remains to consider how far the apprehensions, entertained in many quarters for the future of the British iron manufacture, are justified by actual experience. When we look back upon the past, the growth of British commerce cannot fail to reassure those, who are most inclined to look doubtfully on the future of our industry. Some statistics of the increase in the exports of iron and steel were given in a recent number of 'Iron,' from which I quote the following figures. Our exports of iron in 1840 amounted

to 268,000 tons, of the value of £2,526,000. The quantity in 1850 was 783,000 tons; in 1860, 1,442,000 tons; and in 1870, 2,716,000 tons. The value in the latter year amounted to £21,080,000. In 1872 the quantity was 3,383,000 tons, the value of £36,000,000. We are sometimes assured that Belgium threatens our ironmasters with serious competition; but in Belgium the ore must be carried 100 miles or more to be smelted. The coal-pits are worked in many cases with considerable difficulty, and a Belgian workman does little more than half what an Englishman can accomplish in the same space of time. Sometimes we are told we shall lose our position in the Russian market. The Russian Government are doing their utmost to encourage the manufacture of iron at home; though there is little demand for pig iron in that country. Few Russians have had any experience in puddling. Skilled mill and forge men are scarce. Few of those obtainable have had any experience in the use of mineral fuel, and great difficulty is experienced in consequence of the objection of the Russians to piece-work. Lastly we are threatened with competition from the United States. The production of pig iron in the States may now be estimated at 2,500,000 tons, an increase of 1,000,000 tons on the production five years ago, and yet the ironmasters of the United States, who are protected by a duty of nearly £3 a ton on railroad iron, have hitherto been unable to supply the entire demand at home. There cannot be a doubt as to the ultimate consequences of the comparative exhaustion of our supplies of raw material at home; but we may hope that the tariffs, which now throw obstacles in the way of legitimate trade, will in time be removed, and that, as Mr. Mattieu Williams has suggested, we may be enabled to avail ourselves of the natural resources of America for obtaining our supplies of raw material, just as we already derive large supplies of hematite iron ore from Bilbao.

American competition.

At the present time, the United States, not content with their natural advantages, impose an almost prohibitory tariff on our exportations. There is a party in America opposed to protection, but hitherto the superior organisation and greater determination of the manufacturers interested in the maintenance of the tariffs has overpowered all opposition. At the last annual meeting of the American Institute of Mining Engineers at Philadelphia, the Honourable D. Kelley, who delivered the opening address,

asserted that, by its dereliction of duty in not protecting the labourer of Great Britain against competition, the Government of this country have fostered anarchy in Ireland, while the life of the labourer in England and Scotland has been robbed of all its joys. 'The millions of sturdy men,' he declared, 'represented by Bradlaugh, Odger, Joseph Arch, and the travelled and humane patrician, Sir Charles Dilke, know that the world owes every man a living, and that it is only by protection that the means of living can be secured to the people.' So long as such a feeling prevails, there is little hope of our ironmasters obtaining free access to America.

The progress of the American iron works is the more creditable, because great difficulties are experienced in obtaining a sufficient supply of labour. Men come over from England, having had their expenses paid, on condition of taking an engagement for a period of five years. As soon as their bargain is performed, they generally find it impossible to resist the attractions of an independent farm in the Far West. Their places must be supplied by other workmen, obtained by the same costly means from the mother country. The difficulty of obtaining skilled workmen has had a great effect in America in stimulating the invention of labour-saving machinery; and as scientific manufacturers, the American ironmasters can doubtless hold their own against the world. In finished iron the Americans have been highly successful. Bridge-work, locomotives, wheels and tires, and machinery, are produced at prices, which may compare not unfavourably with our own. As an illustration of American ingenuity and enterprise, which came under my immediate notice, on the occasion of a recent visit to the States, I may point to the Peabody Rifle Company's establishment at Providence, Rhode Island. During the Rebellion the Company was fully employed in the manufacture of small arms. The cessation of the struggle put an end to the demand for rifles; but, with the fertility of resource which distinguishes American industry, the manual skill of a large body of workmen especially apt in the production of tools or machinery, composed of numerous small and interchangeable parts, and the valuable and ingenious plant belonging to the Company, are now employed in the production of sewing machines. Three hundred machines are turned out every day, and the sale is constantly increasing. The wages of the 500 operatives employed are most liberal. The monthly pay-sheet amounts to 25,000 dollars, giving an average of 40s. a week throughout the factory. The leading workmen, five or six in number,

to whom the work is let by the piece, or rather by sub-contract, earn nearly £600 a year. The superior mechanics earn 12s. to 14s.; labourers 4s. to 6s. a day. The supply of highly-skilled labour is limited, but ordinary mechanics can always be obtained. On an average, one skilled mechanic a day makes application for employment.

The success of the Peabody Company affords significant evidence that the cost of production is not augmented in equal proportion to the high rates of pay. At the time of my visit, they were negotiating a contract for the supply of 100,000 rifles to the Roumanian Government, at the rate of 63s. per rifle; and they had to compete for the contract against all the makers of Birmingham and Liège. This Company had also in prospect an order for 200,000 rifles, from the Turkish Government. The success, with which the Americans have reduced the cost of production by the invention of machinery, gives us ground for caution, lest our old supremacy be shaken by the energy and talent of the New World; while it also gives us reason to hope that the effects of the exceptionally high rates of wages now prevailing may be mitigated by substituting, wherever it is possible, mechanical for manual labour.

A monopoly of the iron trade impossible.

I can only repeat once more that, in the present condition of our trade, there is nothing to justify serious misgivings as to our power of continuing a successful competition with foreign producers. It does not follow that, because we have lost a monopoly of a particular branch of trade abroad, the skill of the English workman must have deteriorated, or the cost of production have been unduly enhanced by the rise of wages. Foreign countries may have imported from us a particular commodity at a former time, solely because they were entirely inexperienced in its manufacture. When my father was executing the Rouen and Havre Railway, he imported the rails from England, although he had to pay an import duty at the French Custom-house, amounting to a considerably larger sum than the selling price of the rails at home. The almost incredible difference between the price of English and French rails at that time no longer exists: because that special branch of industry is now as well understood in France as in England. So, too, in the case of the employment of English contractors for

the execution of public works on the Continent. An opportunity was offered to them in the origin of the railway system on the Continent; because in those early days of railways there were no native contractors, sufficiently acquainted with the art of making railways to venture to compete with the English invaders. Their intelligent observation of our methods of construction soon enabled the contractors on the Continent to tender in competition with the English; and for many years past all the railway works in France have been carried out by Frenchmen. It does not follow that the English contractor has lost his former skill. The true inference is, that the French, who had been previously in a position of inferiority solely from lack of experience, were enabled, as soon as they had gained that experience, to execute the works required, without the assistance of foreigners.

The development of our commercial relations with France, since the negotiation of the Treaty of Commerce, affords convincing proof of the great capabilities of our manufacturing industry. Since 1860, the exportation of iron, wrought and unwrought, to France has increased in value £540,000.

Looking therefore to the present condition of our iron trade, there is nothing to justify serious misgivings. According to the last report of the Commissioners of Customs, the average value of the pig iron exported in 1870 was £2. 19s. 2d. per ton; in 1871, £3. 1s. 8d.; in 1872, £5. 0s. 11d.; and yet the demand for pig iron continued unchecked. The increase in the quantity exported in 1872 over 1871 was 28 per cent. The increase in the price ranged as high as 108 per cent.

While the export of pig iron attained to the figures I have quoted, the total increase in the exports of iron and steel manufactures did not exceed 6.7 per cent. Indeed the manufacture of steel actually fell off from a value of £683,000 in 1871, to £623,000 in 1872; a result the more remarkable as compared with the increase in pig iron, because the price of steel had not advanced in the same proportion as the rise in pig iron. The price of the latter article had risen, as I have said, from £3. 1s. 8d. to £5. 0s. 11d. per ton; while unwrought steel had only advanced from £30. 12s. 3d. to £32. 18s. 7d. per ton, and steel manufactures from £52. 8s. 1d. to £55. 4s. 10d. per ton.

Hence it would appear that a demand once created for an article of the first necessity, such as iron, is not easily checked, even by a very marked advance of price.

It must, however, be remembered that, when the course of trade has been changed, and consumers, alarmed by the high prices in our market, have been taught to look for their supplies in another, the position once lost is not easily recovered. The superiority of our artisans in skill and industry has assisted our manufacturers to compete successfully in the past. The same success will not be maintained in future, unless our employers and workmen continue, as before, to use their united efforts to reduce the cost of production.

Shipbuilding.

Perhaps no branch of industry has been more successfully prosecuted in this country than shipbuilding; and the extensive use of iron for ships of the largest type makes it a point of great interest to ascertain how far the cost of building ships has been affected by the recent advance of wages. I am informed by an eminent firm of shipbuilders, that at the close of 1871, shortly after the reduction in the hours of labour from fifty-nine or sixty hours a week to fifty-four, an agitation was commenced amongst all classes of men for an advance in their rates of wages, which has been, in some shape or other, conceded to them, to the extent of from 7½ to 15 per cent. In reality, this was the natural consequence of the reduction in the hours of labour; although, at the outset the leaders of that movement professed that they had no desire to raise the rates of wages.

The reduced hours of labour increased the cost of production of all articles, and led to the necessity for an advance in the rates of wages. In point of fact, the advantage of the reduction in the hours of labour being conceded, on social and moral grounds, the necessity for some corresponding advance in wages followed as a matter of course, and was perhaps not unreasonable. The two causes combined have resulted in an increased cost of production, so far as labour is concerned, of from 20 to 25 per cent. The cost of building first-class steamers and first-class marine engines has, in consequence of the rise in wages and materials, been increased from 30 to 40 per cent.

The actual diminution, by the nine hours' movement, in the amount of work, turned out with a given plant, should, in theory, be only in proportion to the reduced number of the hours of work, or, say, about one-tenth. It is in reality from 15 to 20 per cent.

From an eminent firm on the Clyde, I learn that on riveters' and smiths' piece-work there has been an increase of 20 per cent. and 10 per cent. respectively, in the last two years; on the other hand, in fitters' piece-work there has been a decrease of 10 per cent. The price of first-class steamers in 1871 was about £24 per ton. At present the cost would be from 30 to 35 per cent. higher. While the building of sailing ships decreased in 1871 and 1872, in 1873 there has been an increase in the number built. The building of steamers has not been so brisk in 1873 as in 1871 or 1872; a marked falling-off in orders having taken place since the beginning of this year.

On the Thames I find that piece-work is at least 15 per cent. dearer now than in 1869 and 1870. The operatives, employed in attending to large self-acting machines, which require little manual labour, are only working fifty-four hours instead of sixty hours. Again, there has been a large increase of overtime, since the nine hours' movement commenced. Wages for overtime are higher than for ordinary time. An hour and a half's pay is given for every hour's work, and many men refuse to work unless a certain amount of overtime is given to them.

With these recent reports from shipbuilders it may be useful to compare the general progress of shipbuilding in the United Kingdom, in the last ten years. The tonnage of the ships built increased from 328,000 tons in 1867 to 475,000 tons in 1872. There has been no increase in the registered tonnage in the interval, but the vast increase in the proportion of steam to sailing vessels will fully explain the apparently stationary condition of the mercantile marine, if tested solely by the amount of tonnage. It is equally reassuring to find that, in the estimation of foreigners best qualified to form an opinion, the extent of our merchant navy excites profound admiration. M. Bal, director of the Bureau Veritas, in giving evidence before the French Parliamentary Commission of inquiry into the condition of the French Mercantile Marine, said that to him it seemed almost incredible that England, which has only 27,000,000 inhabitants, had 6,903,000 tons of

shipping, whereas all the other maritime Powers combined had only 6,648,000 tons.

In the United States, until the quite recent, and still but partial, revival of the trade, the decline of shipbuilding had been very remarkable. In a country possessed of less natural resource, the suffering, which would have been entailed on the particular industries, would have been almost insupportable. According to Mr. Wells, 15,000 men were employed in New York, in 1860, in building and repairing marine steam engines. In 1870, fewer than 700 found employment in the same branch of industry.

In France, it would seem, from the report of Mr. West, that a wooden ship costs from £3 to £4 a ton more than a similar ship built in England or Canada; and in regard to iron steamers, the price of wrought iron in France for shipbuilding purposes is so much higher than in England, as to make competition impossible.

Amid the many difficulties of the present time, English employers may perhaps take comfort by looking abroad, where they will generally find that the same problems, with which they have to deal, are presenting themselves, and often in a still more aggravated form.

The engineering trade.

Passing from shipbuilding to engineering, I have ascertained that in an establishment on the largest scale, in which the cost of production has been minimised to the utmost, the increased cost of production in 1871 over 1870 was, for wages, 2.73 per cent., and for materials, 2.59 per cent. Again, the increase in 1872 over 1871 was, for wages, 7.97, and for materials, 7.94 per cent., thus showing that the most liberal application of capital, the most ingenious machinery, and skilful administration, had failed to compensate for the great advance in the rate of wages.

I may also quote the following details from a report received from an engineering establishment with which I am connected.

The average wages of some of the most important trades in our employ in 1871, 1872, and 1873, were as follows:—

Trade.	Year 1871.		Year 1872.		Year 1873.	
	<i>s.</i>	<i>d.</i>	<i>s.</i>	<i>d.</i>	<i>s.</i>	<i>d.</i>
Fitters	29	0	30	0	33	0
Turners	30	0	31	0	34	0
Planers	24	0	25	0	28	0
Slotters	24	0	25	0	28	0
Drillers	20	0	21	0	23	0
Moulders	34	0	34	0	36	0
Dressers	24	0	24	0	26	0
Coppersmiths	32	0	33	0	36	0
Smiths	31	0	32	0	35	0
Strikers	19	0	20	0	22	0
Patternmakers	31	0	33	0	36	0
Joiners	30	0	31	0	34	0
Carpenters	42	0	42	0	42	0
Painters	29	0	29	0	32	0
Platers (boilermakers)	34	0	34	0	36	0
Riveters	28	0	30	0	32	0
Holder-up	24	0	24	0	26	0
Platers (ship yard)	35	0	35	0	36	0
Riveters	30	0	30	0	30	0
Holder-up	23	0	23	0	24	0
Labourers	18	0	18	0	20	0

In reply to my inquiry, as to the effect of the nine hours' movement in diminishing the amount of work turned out, I am informed that, while

wages have considerably advanced, no increased activity on the part of the men has taken place. Indeed, less work is performed in nine hours now than formerly when ten hours constituted an ordinary day's work.

The rise of wages has been very considerable in the last two years. The price of locomotives has, in consequence of these various causes, increased from 25 to 30 per cent. An ordinary passenger engine, which might have been built in 1871 for £2,200, cost in 1872 £2,400, and in the present year the price would be £2,600. In modern marine engines the cost of materials and labour is about equal. An engine, which might have been built in 1871, at £40 per horse-power, would have cost in 1872 £46. In the present year the price has advanced from £55 to £60 per horse-power.

In one of the largest steel and iron works in the North I learn that the wages of skilled hands are now from ten to sixteen shillings a day, and have increased 25 per cent. since 1870.

Lastly, I am informed that there is no appreciable difference in the dress or appearance of the working man in the town, in which my works are situated, that there is more money and more time spent in the public-house, and that time in the morning is not so well kept now as it was before the nine hours' movement commenced. It is suggested to me that the improvement in wages and the shortening of the time came too suddenly upon the working man.

It is sometimes difficult to overcome a feeling of depression as to the future of our mechanical industry. But, when we look to the progress made in the past, there is no ground for discouragement. The value of our exports of steam engines in 1866 was £1,760,000, in 1872 £2,995,000. The value of our exports of machinery of other sorts was, in 1866, £2,998,000; in 1872, £5,606,000. The past has been prosperous, and there is no reason why a cloud should overshadow the future of our industry, if only the time-honoured rule be observed, of giving a fair day's work for a fair day's wages.

Rise of wages on Continent.

I now proceed to examine the situation of affairs among our continental rivals. Valuable materials for such investigation are furnished to our hands by the recently-published reports of our Secretaries of Legation, and by a most important pamphlet prepared by Mr. Redgrave. From these authorities we learn that, in the last ten years, wages at Verviers, a great centre of industry in Belgium, have gradually increased by 20 per cent. and that the working hours are shorter than they were. At Ghent the rate of wages has risen 60 per cent. in the last fifteen years. The average prices of the necessaries of life show an increase in Belgium of 50 per cent. in the last thirty years. Beef and mutton are now 8*d.* per pound, and bread is about 8*d.* the four-pound loaf. The rise of wages has, however, been greater in proportion than the increase in the cost of lodging, clothes, and food.

In Prussia, Mr. Plunkett states that there is a universal tendency to reduce the hours of labour, and to raise the rate of wages. The Breslau Chamber of Commerce state that, in consequence of the increased cost both of labour and raw material, the prices of cotton carded yarn had advanced 10 per cent. on the best and 16 per cent. on the ordinary qualities. In the Silesian cloth trade, in 1871, prices rose 15 per cent.

In the spinning and weaving factories in Silesia, according to a statement by Dr. G. Reichenheim, quoted by Mr. Plunkett, the increase in the rate of wages in the last ten years has been about 30 per cent. for female weavers, while in the case of male labour it is more than double. The same complaints are made, which we hear in this country, as to the effect of higher pay in rendering the operatives less careful in their work, and more insubordinate than formerly.

Low wages and cheap production not convertible terms.

The most recent inquiries tend to establish the fact, which I have, on former occasions, endeavoured to urge on the attention of employers, that underpaid labour is by no means the most economical. It does not follow that, when a workman receives more pay for exactly the same amount of labour, there is no increase in the cost of production. It would be absurd to put such an interpretation on the axiom assumed by my father, when estimating the cost of work, that the cost of labour in a fully peopled country was, as a general rule, the same, whatever might be the nominal

rate of daily wages. But, where the principle of payment by the piece is adopted, (and, trades-union opposition notwithstanding, no other system of payment can be really equitable), there it will be found that labour, when stimulated by a liberal reward, is far more productive than that of the ill-paid operative. The reports to which I have referred are full of illustrations on this point.

Belgium.

In Belgium, all the factory occupiers are of opinion that the English operatives are far superior to the Flemish. An Englishman, being better fed, possesses greater physical power, produces as much work in ten hours as a Fleming in twelve, and, understanding the machinery which he works, he can point to the cause of an accident; whereas in Ghent half-an-hour is constantly lost in seeking for the reason of a stoppage. Although the rates of wages are lower, and the hours of labour longer; English manufacturers have but little to fear from Belgian competition.

Russia.

Mr. Egerton states, that, in Russia, 13 hours a day is the average length of the hours of labour, children generally working the same time as men; and yet there is no country in which there is so great a waste of labour. In mills where the best and newest machinery is used, it is necessary to limit the earnings, which, if large in amount, would be expended in drinking. In England a spinner will, with his assistants, attend to 2,000 spindles. In Russia, he never has more than 1,000, and generally 500 spindles under his charge.

Switzerland.

Mr. Gosling says of the Swiss workman, that he is inferior to the British workman in physical strength and energy.

France.

The French manufacturers insist strongly on the greater cost of production in their country as compared with England. They estimate the cost of wages per week for the hands employed upon 10,000 spindles at £59. 10s., as compared with £41, which would be the corresponding amount in an English factory. 'The value of the English workman,' says Mr. Redgrave, 'still remains pre-eminent, although the interval between him and his competitors is not so great as it was; he has not retrograded, but they have advanced.' We see too much of intemperance in England, but there is much reason to complain in Belgium and the manufacturing districts of France, where the cheapness of intoxicating liquors is a fearful temptation to the working classes.

Increasing production with the same number of hands.

The progressive development in the skill of our factory operatives has been clearly shown in the comparison, instituted by Messrs. Bridges and Holmes, of the tasks, now performed, with the amount of work allotted to the hands, as ascertained by the Factory Commission of 1833. Messrs. Bridges and Holmes estimate that the proportion of spindles in 1833 was 112 to each hand, while the corresponding number at the present day would be 517 spindles. The speed of the mule has been so much increased, that more stretches are now made in 10½ hours than formerly in twelve. In 1848 a female would have had only two looms, now she will attend to four. The speed of the power looms in 1833 varied between 90 and 112; it now varies between 170 and 200 picks a minute. Notwithstanding all the improvements of mechanism, the cotton-weaver of the present day is subject to a greater strain than his predecessor of forty years ago.

From a consideration of all these facts, we have reason to congratulate employers in England on the possession of a body of workmen superior to those of any other country. We may also assert, on their behalf, that in no other country of the Old World is the same solicitude displayed for the welfare of the workmen.

I observe with regret the frequently repeated manifestations of disaffection on the part of the working classes on the Continent towards their employers. Lord Brabazon, in his able report on the condition of the industrial classes in France, quotes some painful illustrations of the entire

want of confidence between class and class in that distracted country, where 'Communistic principles have done so much to alienate the affections of the workmen from their employers,' and where a large proportion of those engaged in manufacturing industry live in a condition of wretchedness and misery, of which, I venture to hope, very few of those, who can command regular employment in this country, have any experience. At Elbœuf we are told of a certain manufacturer who, during the period of dearth, bought a large quantity of provisions, with the view of reselling them to his workmen at a low rate, but who was obliged to renounce his humane project; because the workpeople imagined it was a pretext for making money out of their misery. At Lyons, where no social distinctions keep asunder the numerous small employers from the employed, the sympathy which formerly existed between the owner of the loom and his assistants is no longer found.

When I turn from this gloomy picture to those bright recollections, the most precious portion of the heritage, which I have received from my lamented father, and call to mind the cordial relations, which he always preserved with vast multitudes of workmen, and with a large staff of agents of every grade and disposition of mind; still more, when I see among contemporary employers so many evidences of the same success in conciliating their dependents, I thank my God from the very bottom of my heart, that I was born an Englishman.

Employment of children in Belgium.

While in England we are happily doing away with the great evil of employing young children in our factories, all the Chambers of Commerce in Belgium unite in deploring the increasing moral and physical degeneracy of the working classes, owing to the premature employment of children. In the Belgian factories for spinning and weaving flax, cotton, and wool, children from ten to twelve years old are very generally admitted, and work twelve hours a day. In the Belgian coal-pits 8,000 children under fourteen years of age, of both sexes, are employed. Of children between ten and twelve 2,400 are employed, 700 above and 1,700 below ground.

In 1866, out of their total population of 4,827,000, more than one-half were unable to read or write. The necessity for the employment of children is best proved by the description given by Mr. Kennedy, of the position of the

Belgian operatives at Alost and Tirmonde, where a first-class hand earns £28 a year, while the smallest sum on which a man can exist is £20 a year. Indeed, existence is only made possible by the employment of children in factories, and by the possession of a small garden in which vegetables are raised.

Employment of women in England.

In the English factories, where a larger proportion of women are employed than in the factories abroad, it has recently been proposed that the number of hours of labour should be limited by law. The proposal is supported by Messrs. Bridges and Holmes, on the ground that, by exciting a spirit of rivalry between them, women can be goaded on to over-exert themselves in a manner, which would not be observed among men. A woman, we are told, who can mind four looms without an assistant has a certain position, and becomes an object of attention. 'Hoo's a four-loomer; hoo's like to be wed,' will be commonly remarked of such an one.

The Association of Employers, though differing on almost every other subject from Messrs. Bridges and Holmes, suggest that women should be excluded from factories for three months after their confinement. Great evils have been found, by experience, to ensue from the too early return of the mothers to factory labour. Let us venture to hope that another Session of Parliament will not be allowed to pass by, without placing on the Statute Book a legal prohibition against a practice, which is universally condemned by those most competent to form an opinion.

Shorter hours of labour.

The demand for a reduction of the hours of labour, which has been so strongly and successfully urged by certain classes of our operatives, is not universally supported either at home or abroad.

The average length of a working day in Switzerland is twelve hours, exclusive of the time for meals. The general tendency is to a reduction of hours, and laws have been passed, limiting the length of the working day in some cantons to twelve hours. These changes are, however, almost entirely due to the efforts of local politicians. A proposal of this nature recently

made in Zurich, and sanctioned by the Cantonal Legislature, was eventually thrown out by the popular vote.

At Rouen, Mr. Redgrave found no strong desire for a diminution of the hours of labour in the cotton factories. The operatives were chiefly solicitous for a rise of wages. On this subject the workpeople of all countries seem to entertain similar views. Messrs. Bridges and Holmes, in their report on the condition of operatives in English factories, say that the workpeople are by no means unanimous. Among the women especially, many are apathetic, and some are positively opposed to a limitation of the working hours.

In the United States, at Lowell, near Boston, I ascertained by personal inquiry on the spot, that the working hours were sixty per week, and that no indication had yet been given of a disposition among the operatives to reduce the hours.

Decline of pauperism.

Though there may be reason to regret that the working class have not reaped more substantial and universal benefits from the recent additions to their wages; we have at least the satisfaction of knowing that there has been some slight decrease in the amount of pauperism, and in the proportion of persons in receipt of relief to the whole population.

Emigration.

Meanwhile the tide of emigration has never ceased to flow. The proportion, too, of English and Scotch to Irish has, of late years, much increased. In 1872 the wide labour field of the United States absorbed 199,000 out of a total number of 252,000 emigrants from the United Kingdom. In the same period 33,000 sailed for the ports of British North America. The United States offer many advantages to the newly-arrived emigrants. The homestead law gives, for a merely nominal sum, the right to a homestead and 160 acres of land to every individual, who is actually a citizen of the United States, or has declared his intention of becoming such.

In examining the various circumstances, which tend to raise the price of labour in England, the prevailing high wages in the United States, and the increased facilities for emigration, must ever be kept in view. The nominal rate of wages in America may indeed frequently convey a delusive idea of prosperity; yet it cannot be doubted that the thrifty, skilful, and industrious artisan has large opportunities of advancement in the New World.

A great majority of the emigrants go out to join some friends already satisfactorily established. When this is not the case, it is essential to the emigrant's success that he should have accumulated not merely a sufficient sum, to defray the cost of his voyage across the Atlantic, but enough to enable him to travel, if necessary, far into the interior, and to visit, it may be several, rapidly rising cities in the West, before finally settling down. The artisan, who is able to maintain himself for some months after landing at New York, and to make a wide exploration of the country, will be sure in the end to find a favourable opening. Alas, how few of those who emigrate from this country are possessed of such resources!

Wages in America.

Many examples of the prosperity of the working classes came under my own observation on a recent visit to America. The workpeople are paid as far as possible by the piece. The monthly pay-sheet at the Merrimac Mills, at Lowell, where 2,600 hands are employed, amounts to 75,000 dollars, which gives an average of thirty dollars a month, or 30s. a week. The majority of the workpeople are Americans, but there are many from Canada and the Old Country. The proprietors of the mills have established several lodging-houses for the unmarried women whom they employ. At each of these houses some thirty women are lodged. The house is placed under the supervision of a respectable matron. The cost of living is 3½ dollars a week, and female operatives can earn from 14s. to 16s. a week over and above the cost of their board and lodging. The men pay for board 2s. a day, and their wages vary from 7s. to 10s. a day.

At the Lonsdale Company's Cotton Mills, near Providence, in a factory containing 40,000 spindles, one spinner attends to 1,408 spindles, and in weaving, one weaver attends to from four to six looms. In England, the proportion would be, on the average, one hand to every three looms,

working at a higher speed than they have attained in America. Male weavers were earning from 44s. to 52s., and female weavers from 40s. to 44s. weekly. Spinners earn from 4s. to 6s. a day. Women pay for board and lodging in lodging-houses, provided by their employers, 12s., and men 16s. a week. The operatives, earning these wages, are better able to save money than the operatives in our own country; and many of the hands at the Lonsdale Mills have £1,000 to their credit in the Savings Bank. At the great Harmony Mills at Cohoes, near Albany, where 4,000 hands are employed, two-thirds are emigrants to the States, principally English and Scotch, although there are many Germans and some French. The general wages are for women from 3s. to 6s. a day, for men from 6s. to 10s. a day. The cost of living is moderate, and assuming that a female operative earns 28s. a week—by no means a high average—she has 16s. a week to spend on dress and luxuries. At Cohoes a weaver attends to four, five, or six looms, but the machinery is not worked at so high a rate of speed as in Lancashire. The mule is never worked at a speed exceeding three stretches a minute.

In Quebec wages have of late been rapidly advancing. Artisans can now command 8s. a day, and labourers employed in unloading ships, whose employment, however, is uncertain in summer, and in winter wholly ceases, earn 10s. to 12s. a day. A man with a family can live well on 4s. a day. The long winter is the great drawback to the prosperity of the working class in Canada. Quebec has its Wapping, its extensive suburbs, chiefly occupied by the working classes; and there is no external indication in these quarters of a condition of life superior to that attained by the majority of our working men at home. In the Ottawa district, in Canada, the young farmers are able to find employment in winter by leaving their homes, and going up to the forests to cut timber. They earn 30s. a week, and they are boarded in addition. In the spring the lumberer returns home with a considerable sum of money saved. He carries on his farming operations throughout the open season, and returns to the forests in the autumn. The life is toilsome, and involves a long separation from the fireside at home; but the perseverance of a few years will result in the accumulation of a valuable capital for farming operations, and secure to the settler his future independence.

Ottawa is one of the rising towns of Canada. Its prosperity is derived from the timber trade, and from its being the seat of the Government. Wages in Ottawa were last year (I speak of 1872) extravagantly high. Masons were

earning 14s. a day. All classes of artisans employed in building were paid from 10s. to 12s. a day. For four or five months in winter building operations are suspended; but provisions are cheap, and house rent is the only costly item.

At Hamilton, in Upper Canada, the wages for artisans are 8s. a day. House-rent is about 28s. a month. The expense of fuel in winter is nearly equal to the sum paid for house-rent. Food is cheap. A stock of salt beef can be laid down for the winter at the price of $1\frac{1}{2}d.$ a pound. The agriculturists in the Hamilton district are in a prosperous condition. Every settler travels in a light waggon, drawn by a pair of serviceable horses. The population seemed robust and healthy.

In other settlements forming part of the Dominion, the appearance of the people was less satisfactory. At Charlottetown, in Prince Edward's Island, the universal vehicle is drawn by one horse instead of two, as at Hamilton. It was sad to see the population generally so pale and thin, and, in appearance, sickly and out of health. It is hard to find a reason for this marked physical deterioration of the descendants of Scotch, Irish, and English settlers. Probably the long winter is, to a great extent, the cause. The impossibility of active outdoor operations at that season, and the consequent temptation to spend the day in smoking and drinking in overheated rooms, is extremely prejudicial to the health of the population. At Picton, in Nova Scotia, the inhabitants appeared more robust. The wages for ordinary shipwrights are 8s. a day, and taking into consideration the cost of living, the working classes are as well off as in any part of Canada.

In comparing the American and English operatives, or, rather, the English operatives, when transplanted to the States, with the hands who have remained in the Old Country, it would seem that there is, as a rule, a higher development of skill in the individual operatives. The difference is attributable to the conviction that the present high rate of wages in the States could not be maintained; unless the utmost skill and diligence were put forth.

Arbitration.

The results which have followed from the reference of disputes relating to wages to arbitration are a sign of the happiest augury for the future relations between employers and employed. It has been urged, on the part of the employers, that the working class will only accept the decision of arbitrators, when it is favourable to themselves. But in this, as in many other respects, the organisation of the trades unions, and the influence which the more enlightened workmen, acting as members of the executive committees of the unions, possess over their less-instructed fellow-workmen, have been the means of securing obedience to every decision arrived at after careful investigation, conducted in an impartial spirit. Such influence becomes more important when the members of the trades unions are for the most part uneducated men. It is always more difficult for an employer to negotiate or to argue with a boiler-maker than with a fitter. The executive councils of the unions have entitled themselves to the gratitude of the employers of labour, by accepting the use of machinery, the substitution of which for manual labour becomes more and more indispensable with every advance in the standard of wages.

Legitimate province of the trades unions.

It is not by encouraging useless strikes, or by making an attempt, which in the end must always be defeated, to sustain a vast body of workmen and their families, when not in the receipt of wages, that the wire-pullers of the trades unions will best serve the interests of their clients, or enhance their personal influence among them. But there is a wide field of usefulness open to these captains of our great hosts of workmen, in which success is to be attained, not by war, but by diplomacy. The state of the trades, in which their clients are employed, should be carefully watched, and every variation in the prices quoted, every fluctuation in the cost of the raw materials should be noted. And here I may frankly admit that the proposal of the International for a universal strike contained a few grains of wisdom; for it is clear that, if the cost of producing an article in England were so much enhanced by an advance of wages, that the foreign manufacturer would be enabled to undersell us in every market, it would be an act of self-destruction for English workmen to insist upon a rise, which would have the inevitable effect of depriving them of employment.

In such a case, unless the workmen in the competing countries can agree to act in concert, an advance is impossible; unless by superior skill or machinery the more highly paid workman is able to turn out a larger amount of work.

Foreign competition.

It has been already pointed out that in England we have to contend against competition of two kinds—against the cheaper labour of the Continent on the one side, and against the superior natural resources of America on the other. While we occupy at the present time a highly favoured position, which has been attained not merely by the skill of our workmen, but by the administrative skill of their employers, and the gradual accumulation of an ample capital in their hands, the race with other great manufacturing countries is very close. The Swiss have entered into competition with our own manufacturers, both in the home and foreign trades. The exports of textile fabrics from Switzerland, as we learn from Mr. Gosling's report, have risen from £12,485,000 in 1860, to £26,464,000 in 1871, an advance of 112½ per cent. In this total the exports to the United States have risen from £509,000 in 1862, to £2,159,000 in 1872, in other words, over 324 per cent. In cheap silks and ribbons the Swiss are able to compete with the British producer in the English market; and, to sum up the case in the words of Mr. Gosling, 'The advantages of Switzerland in competition with Great Britain are the use of water power as a substitute for steam power to the extent of upwards of 80 per cent., low wages, long hours of labour, and a minimum expenditure for management.' On the other hand, as an inland country, Switzerland has to pay heavy freights, the workmen are inferior in activity to our own, buildings for machinery are more costly, and from want of capital, production is on a smaller scale than here. The balance, however, seems to be greatly in favour of Switzerland, and cannot fail to become greater from day to day.

Such being the case as regards textile industry, Mr. Lothian Bell has recently pointed out, that, in ores of the finer descriptions, the resources of the United States are unlimited, while in coal our own wealth is, in comparison, poverty. There is but one bar to the boundless production of

minerals in the New World, that is to say, the want of hands to manufacture them.

Communism in Germany.

A large number of the working class in Germany have been fascinated by the fanciful theories of Lassalle. His system is founded entirely upon the pernicious principle that the State is to do everything, and the people nothing for themselves. Karl Marx, as the successor of Lassalle, is the ruling spirit of the German socialists, and has become a prominent figure from his connection with the International. The socialist journals in Germany delight to reproduce the programme and doctrines of that society. They make noisy professions of atheism. They applauded the insurrection of the Commune in Paris. They have a collection of songs of their own. They disavow the warlike policy of Germany, and have endeavoured to substitute the community of class interests for the community of race, language, and country. It must not, however, be supposed that the number of these unpractical visionaries is proportionate to the noise, which they make in the world. The influence of socialistic doctrines is not so great in England as on the Continent, and it is weaker in America than in England. I hope, therefore, that no disposition may be manifested here to abandon the hopeful work of social, moral, and material progress for the pursuit of visionary and impossible schemes.

The amelioration of the condition of the poor is not to be brought about by destroying the ancient fabric and foundations of our social and political system. It is easy to destroy but most difficult to restore the institutions created by past generations, in which there lived men not less great, and wise, and good than the most gifted of our own contemporaries. Mr. Ruskin, a devoted friend of the working classes, in a passage of more than ordinary eloquence, has truly said, 'This is the thing, which I know, and which, if you labour faithfully, you shall know also,—that in Reverence is the chief power and joy of life;—Reverence, for what is pure and bright in your own youth; for what is true and tried in the age of others; for all that is gracious among the living, great among the dead, and marvellous in the Powers that cannot die.'

Co-operation.

Our artisans may believe that the profits of former days were so large that employers can afford to pay the present rates of wages, without raising their charges to the consumers. There is but one means, by which this fallacy can be exposed. The workmen must become to a certain extent their own employers. In a co-operative establishment, created in part by his own hardly-earned savings, the handicraftsman will find himself called upon to apportion equitably the earnings of his business between labour and capital. In this double relation he will learn how great are the difficulties, which beset the employment of capital in productive industry in a country, in which competition is so keen as it is in England. In no other country does capital command so low a rate of interest; and, if large accumulations of capital have been made, and money is therefore cheap, it should not be inferred that the rate of profit has been high by comparison with other countries, but rather that our employers of labour, as a class, have been distinguished by their frugality, their perseverance, and their enterprise. I am grateful, therefore, to Mr. Holyoake, for his vindication of their claims at the recent Co-operative Congress. He justly said that capital was the enemy of nobody, but rather the nursing mother of production.

The co-operative principle, in its application to the business of distribution, has been already most successfully developed. My desire is to encourage working men to create co-operative establishments for the purposes of production. The accumulation of the necessary capital is an obvious difficulty; but as wages were never so high as at present, so this obstacle can be more easily surmounted now than at any former time. Some men may object to recognise the special responsibilities of a fellow-workman holding the office of manager of a large business, by giving a proportionate salary. It is because the recognition of authority is essential, whenever anything practical is to be done, that the International Society has shown such uncompromising hostility to the co-operative principle. The denial of a proportionate reward for superior intelligence or industry is the first article of its catechism. The absurdity of attempting to combine the energies of the men for any definite object, without placing a competent chief at their head, has been humorously exposed by Mr. Carlyle. 'Ships,' he said, 'did not use the ballot-box at all, and they rejected the phantasm

species of captains. Phantasm captains with unanimous votings! These are considered to be all the law and prophets at present. If a man shake out of his mind the universal noise of political doctors in this generation, and in the last generation or two, and consider the matter face to face with his own sincere intelligence looking at it, I venture to say he would find this a very extraordinary method of navigating, whether in the Straits of Magellan, or the undiscovered sea of time.'

English workmen are less easily deluded by tall talk and sophistry than the more excitable populations of the Latin race; and I would earnestly invite them to apply their practical sagacity to the difficult yet hopeful experiment of co-operative industry.

Demand and supply determine wages.

I must once more repeat the familiar axiom, that the price of labour, like that of every other commodity, must mainly depend upon the relation between supply and demand. The wages of skilled workmen have risen, because skilled workmen are scarce. How shall we increase their number, and improve their skill? My answer is, by bringing recruits into our industrial army from a class of society, which has hitherto exhibited too strong a prejudice against manual labour. The same aversion to handicraft of every kind exists in the United States and Canada. In America a skilled workman earns 30 dollars, a clerk only 15 dollars a week; and, while it is almost as difficult for a clerk to obtain a situation in New York as in London, a skilled workman can always command employment. It is unnecessary to dwell on the evils that must ensue from a disproportionate increase in the non-productive classes of the community. Lord Bacon has truly said, that a population is not to be reckoned only by numbers, for a smaller number that spend more and earn less do wear out a greater number that live lower and get more. My father's advice was often sought by parents anxious for the future of their sons. His counsel always was, that a young man, whose destiny it must be to make his way, unaided, through the world, should begin by learning a trade. It is a laudable ambition in a parent to endeavour to raise his family to a better station in life. He cannot bestow on his children too high an education. But a wise man will be on his guard, lest the enjoyment of such advantages should render those

occupations distasteful, which afford the most secure and ample livelihood to those whose lot it is to labour. When justly appreciated, the condition of the skilled artisan should be as much esteemed as that of any other class of the community. He whose life is passed in performing such needed services for his fellow-men, whatever his special calling, holds an honorable station, and social dignity will ever be most effectually maintained by those who are the least dependent upon the favours of others.

Forethought.

In conclusion, I would tender a few words of advice to my fellow-countrymen of the so-called working classes, for whose welfare I am bound to feel the deepest solicitude. Their just claim to share in the benefits arising from a thriving industry has of late been liberally recognised. The earnings in many trades have been unprecedented. It should not be forgotten that forethought is an especial duty in a time of prosperity. At no distant period, the progress of our commerce may sustain at least a temporary check. It will be sad indeed if the receding tide leaves behind it multitudes of our highly-paid workmen without the slightest provision to meet a period of adversity.

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